

AbroadShield AI — Technical Build Brief

Stack validation, build sequence, cost/time reality check, and uncovered risks

Prepared July 25, 2026 | Research grounded in web verification where noted | Judgment calls are labeled explicitly

This brief evaluates your locked stack, fills the gaps you asked about (hosting, monitoring, OCR, WhatsApp routing), and gives a grounded build sequence and cost/time estimate for a solo, AI-assisted builder. Every factual claim about a tool, price, or policy below was checked against current sources (dated 2026) rather than pulled from memory. Anywhere I'm giving an opinion instead of a verified fact, it's marked **JUDGMENT**, not fact.

1. Is Your Locked Stack Still the Right Call?

Short answer: yes, with two things to watch closely and one gap that's more dangerous than it looks.

LangGraph — confirmed, still the right pick

LangGraph and LangChain both reached 1.0 LTS status in October 2025, and LangGraph is the current production standard specifically for stateful, cyclic, multi-step agent workflows — which is exactly your milestone-sequencing use case. LangChain's older AgentExecutor pattern is deprecated (maintenance mode until December 2026), so building on LangGraph directly, as you've locked in, avoids that dead end. The tradeoff the market consistently flags: a real learning curve around state schemas and checkpointers. For a solo AI-assisted builder this means budgeting extra time for the state machine specifically, not treating it as a thin wrapper you'll finish in an afternoon.

FastAPI + Supabase — confirmed, still correct for MVP scale

No change needed. Both remain standard, well-supported choices for this shape of product. Two things about Supabase's free tier specifically matter for you, because your existing CWM backend already lives on a similar pattern:

- Free-tier Supabase projects auto-pause after 7 days of no API activity, and a paused project takes 20–30 seconds to wake up on the next request. That's fine in development. It is a bad look the moment you demo this to a survey respondent or an investor and the app stalls on load.
- Free tier caps: 500MB database, 1GB file storage, 50,000 monthly active users, 5GB egress, max 2 active projects. For a Phase-1 pilot with dozens to low hundreds of real users, this is genuinely enough — you do not need to pay for Supabase on day one.
- The jump to Pro (\$25/month, removes pausing, adds backups) is the trigger to pull once you move from private testing to anything you're sending real students to on a recurring basis — not before.

AiSensy — confirmed, but the pricing model changed under you

AiSensy is still a legitimate, India-based Meta-official WhatsApp Business Solution Provider, and it still gives you free API access with a Free-Forever plan. But Meta changed how WhatsApp charges as of January 1, 2026: billing moved from per-24-hour-conversation to per-template-message-sent. For India specifically, that's roughly ■1.09 per Marketing-category template and ■0.145 per Utility/Authentication-category template. Messages you send inside an active 24-hour customer-service window (i.e. replies to something the student just sent you) remain free regardless of category. This is a materially better pricing shape for you than it would have been under the old model, because your core loop — proactive deadline nudges — are Utility-category messages, the cheapest tier by a wide margin. But it also means template *category correctness* is now a direct cost lever, not just a compliance checkbox (more on this in Section 4 — this is the sharpest edge in your whole architecture).

The gap you asked about: document parsing / OCR

This wasn't in your locked stack and it needs to be, because Pre-Departure gap-checking is one of your four core differentiators and it is entirely dependent on being able to read a photo of a passport, visa page, or bank statement taken on a mid-range Android phone in a tier-2 town — not a clean scanned PDF.

Option	What it's good for	Cost shape (verified 2026)
LlamaParse (LlamaIndex)	Agentic OCR with semantic structure output (Markdown/JSON), strong handwriting support, designed for feeding LLM pipelines rather than raw text dumps. Free tier (10k credits) is enough to prototype Phase 1 fully before you spend anything.	Free tier to start; usage-based tiers (Fast / Cost-Effective / Agentic / Agentic Plus) after.
Google Document AI	Purpose-built identity-document processors (passport, ID) with high accuracy, enterprise-grade. Better suited once you have volume and want a managed processor per document type.	~\$0.10 per identity document processed, plus processor hosting (~\$0.05/hr per deployed processor version, roughly \$438/yr if left running).
Mistral OCR / AWS Textract / Azure Doc Intelligence	Cloud-native alternatives if you standardize on that cloud elsewhere. All support handwriting and structured field extraction.	Per-page pricing, roughly \$0.001–\$0.06/page depending on provider and tier.

JUDGMENT: for a solo builder validating Phase 1, start with LlamaParse. It's the lowest-friction integration, the free credits cover your entire pilot, and its structured output plugs directly into a LangGraph node without extra glue code. Move to Google Document AI or a cloud-native option only once you have real volume and want per-document-type tuning — not before.

The gap you didn't ask about but need: hosting and observability

- **Hosting:** You're already on Render for CWM's backend, so reuse that experience. Render's free Hobby tier still exists in 2026 but spins your service down after 15 minutes of inactivity (1-minute cold-start on next request) and caps you at 750 hours/month and 5GB bandwidth (cut from 100GB in an April 2026 revision). For an MVP that real students rely on for deadline nudges, the free tier's sleep behavior directly undermines your core promise ("the AI that doesn't disappear") — budget for the Starter tier (~\$7/month per service) from day one rather than free.
- **LLM observability:** Not in your locked list and worth adding — Langfuse (open-source, generous free self-host or hosted tier) is the current standard companion to LangGraph for tracing agent steps, prompt versions, and token cost per student journey. You'll want this before Phase 1 goes to real users, not after something breaks in production and you have no trace to debug from.
- **Error monitoring:** You already run Sentry on CWM. Reuse the same account/integration pattern for AbroadShield's FastAPI backend rather than standing up something new.

2. Realistic Build Sequence for a Solo, AI-Assisted Builder

This follows your own locked verdict — boring infrastructure before personality — but breaks it into an order that lets you validate assumptions before you've sunk weeks into the wrong phase.

Step 0 — Resolve the survey question first

Before writing a line of backend code: confirm whether your 13-question Google Form is actually live and collecting responses, or whether that's aspirational language in the brief. Everything below assumes real answers exist to steer scope. If the form isn't live, that is today's actual Step 0, not the infrastructure.

Step 1 — Supabase schema + milestone-template table

Build the schema you've already identified as the highest-value proprietary asset: destination country → phase → required steps. Populate it manually for one country first (pick the one your survey respondents are actually going to, not a hypothetical). Do not try to cover multiple countries before one is fully correct — a wrong requirement is worse than a missing one on something with visa-rejection stakes.

Step 2 — WhatsApp Business API setup + compliant opt-in flow

Register the WhatsApp Business Account through AiSensy, get Meta business verification started (this can take days, sometimes longer if documents get kicked back — start it in parallel with Step 1, not after). Build the opt-in mechanism (explicit consent capture, clear category of what they're signing up for, working opt-out) before you send a single proactive message. This is not optional groundwork — see Section 4.

Step 3 — LangGraph state machine for Phase 1 only

One phase, one country, one milestone-template. Document upload → gap-check against LlamaParse output → deadline tracking → Utility-template nudge. Resist the pull to wire in Phase 2–4 logic now, even though the four-phase vision is the whole pitch — an unfinished four-phase skeleton is worth less than one phase that actually works for a real student.

Step 4 — Document gap-checking via LlamaParse

Wire LlamaParse's structured output into a LangGraph node that checks extracted fields against the milestone-template's required-document list. Build a human-in-the-loop fallback for low-confidence extractions from day one — a photo of a passport taken at an angle on a low-end phone will sometimes come back ambiguous, and this is the one place in the product where a wrong automated answer does real damage.

Step 5 — Outbound nudge engine

The proactive-nudge pillar — your strongest differentiator on paper — is also the one with the most policy surface area. Build the Utility-template nudge logic last, after opt-in and categorization are solid, so you're not sending real messages against an unfinished compliance layer.

Step 6 — Test with real survey respondents, one phase, one country

This is the point where your own "Immediate Next Steps" from the brief actually becomes actionable — you cannot meaningfully test with respondents until Steps 1–5 exist for at least one real country and one real phase.

Notice what's absent from this sequence: the CV-tailoring, job-matching, and networking/alumni-outreach features from Phase 4. Nothing about your architecture requires deciding those now, and building them before Phase 1 works for even one real student would repeat the exact pattern already flagged in Duby.AI and PUSTAK.AI — building bigger frameworks before executing the smallest revenue-or-validation-generating step.

3. Time and Cost — Grounded, Not Optimistic

JUDGMENT — this entire section is my assessment based on typical solo/AI-assisted build velocity and the scope you've locked, not a verified external benchmark. Treat it as a planning range, not a guarantee.

Time to a real, testable Phase-1 MVP

Workstream	Realistic solo timeline
Schema + milestone-template (one country)	3–7 days
WhatsApp Business verification + opt-in flow	1–3 weeks (verification delay is largely out of your control; build the opt-in UI in parallel)
LangGraph Phase-1 state machine	2–3 weeks (this is where the real engineering time goes — budget more here than anywhere else)
OCR/gap-check integration	1 week if using LlamaParse directly; longer if you build custom confidence-routing
Nudge engine + template approval cycles	1–2 weeks (Meta template approval typically runs ~24 hours per submission but rejections and resubmissions are common for a new WABA)
Buffer for the thing that always breaks	1–2 weeks

Total: realistically **6–10 weeks** of part-time, AI-assisted solo work to a Phase-1 MVP that is actually safe to put in front of real survey respondents — not a weekend, and not the 2–3 days a marketing page for a no-code tool might imply. Your own CWM mobile app timeline (still carrying open P1 bugs after sustained work) is a more honest reference point than any generic "build an MVP in a weekend" claim.

Monthly infrastructure cost during pilot (dozens to low hundreds of users)

Line item	Estimated monthly cost
Render (Starter tier, 1–2 services, avoid free-tier sleep)	\$7–\$15
Supabase (Free tier is sufficient at this scale)	\$0 (upgrade to \$25 Pro once pausing risk is unacceptable)
WhatsApp template costs (India rates, Utility-heavy usage)	Roughly \$5–\$25, dominated by Utility templates at $\blacksquare 0.145$ each — very low per-message cost at pilot volume
LlamaParse OCR	\$0 during free-credit period; low usage-based cost after
Claude/Grok API usage (drafting + conversational)	\$10–\$40, usage-dependent — track this directly since it scales with message volume, not user count
Langfuse (observability)	\$0 on free/self-hosted tier at this scale

Realistic total: roughly \$25–\$100/month to run a genuine pilot, not thousands. The dominant future cost driver as you scale won't be infrastructure — it will be WhatsApp Marketing-category messages if you ever add promotional outreach, and LLM token usage if conversational drafting volume grows faster than your user count.

4. Risks and Blockers Not Yet Covered

1. WhatsApp opt-in compliance is the sharpest edge in this entire architecture

Your strongest differentiator on paper — proactive outbound nudging — is also Meta's most heavily policed behavior category. Verified 2026 policy: any business-initiated message sent outside an active 24-hour customer-service window requires documented, explicit opt-in (a ticked checkbox or an inbound keyword trigger — pulling numbers from a form submission is *not* sufficient consent on its own unless the form itself captured explicit WhatsApp opt-in language). Templates must be correctly categorized — a Utility template used to smuggle in promotional content is a common cause of template bans, and repeated violations escalate from throttling to permanent, non-appealable WABA number bans with no ability to re-register that number. Practically: build the opt-in capture into your very first onboarding screen, not as a later compliance patch, and keep marketing-category messaging entirely separate from the utility-category deadline nudges that are your core value proposition.

2. Meta Business Verification is a real dependency, not a checkbox

To get a verified WhatsApp Business Account you need a registered business entity behind it. You have CWM's OPC registration already — confirm explicitly whether AbroadShield needs its own registered entity or can legitimately operate under an existing one before you hit this step, since restructuring it after WABA approval is more painful than deciding it up front.

3. You're storing some of the most sensitive documents a person owns

Passport scans, visa documents, bank statements — this is high-stakes PII, and a meaningful share of your users will be minors (17-year-olds heading abroad for undergrad, for instance) or their parents acting on their behalf. India's Digital Personal Data Protection Act is in force and applies to exactly this kind of data. This isn't a reason to not build it — it's a reason to put real access controls and encryption-at-rest on the Supabase document vault from the first line of schema, not as a Phase-2 hardening pass.

4. OCR will be wrong sometimes, and this product cannot afford a silent wrong answer

A missed or misread visa document is, by your own brief, capable of ending years of preparation instantly. Any OCR provider you pick — including the best ones — will occasionally misread a rotated, poorly lit, or low-resolution phone photo of a real-world document. The gap-checking node needs a confidence threshold that routes uncertain extractions to "please re-upload" or a flagged manual review, not a confident wrong answer presented as a pass.

5. Free-tier infrastructure quietly contradicts your core brand promise

Your entire pitch is built on "the one relationship that never disappears." A Render free-tier service that goes to sleep and takes a minute to wake up, or a paused Supabase project on a demo link, undercuts that promise at the exact moment someone is deciding whether to trust you with their visa documents. This is a small monthly cost (~\$7–\$15) to remove a real risk to first impressions.

6. LLM choice affects reliability of the state machine, not just conversation quality

LangGraph nodes that decide "what milestone comes next" need consistent, structured output, not free-form prose. Whichever of Claude or Grok you use for drafting/conversation, make sure the milestone-sequencing and gap-checking nodes use structured/tool-call output rather than parsing free text — this is a common, avoidable source of state-machine bugs in exactly this kind of LangGraph build.

Bottom line: your locked stack survives scrutiny with two adjustments (paid-tier hosting from day one, LlamaParse added for document parsing) and one hard requirement that isn't optional — a compliant WhatsApp opt-in and template-categorization system built before Step 5, not around it. Everything else in this brief is sequencing and cost realism around a stack that was, on the whole, a sound set of choices.